tooling.md

> **CivicSync · Tooling and Integrations** Version 1.0-rc1 · 28 Oct 2026 · Owner: Team Leader (Aayush) · Co-owners: Backend Lead (Manindra Shukla), Flutter Lead (Ashmit Singh)
> **Read this first:** prices, free-tier limits and licences change. Items marked **VERIFY** are from memory or from third-party summaries and must be checked on the official page *the day you adopt them*. Pricing facts below were checked against public pages in October 2026.

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1. How to read this file

Every tool has four things: **what it's for**, **why it fits CivicSync**, **cost or licence catch**, and the **sprint** where it first matters. Tool types are **MCP** (a connector that lets an AI coding assistant use a service), **IDE** (editor plugin), and **OSS** (open-source library or repository).

Selection rules: (1) free or inside the budget; (2) permissive licence where possible; (3) actively maintained; (4) the team can learn it in a day. Copyleft licences (AGPL/GPL) and anything that needs a paid seat are flagged.

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2. Budget: the under-$50/month pilot plan

This is also the "itemized BOM and cost" for the DTU–SEVA dossier (point 5). CivicSync is software-only, so there is no hardware BOM; test phones are the team's own.

Item	Plan	Expected monthly cost	Note
Supabase (Postgres + PostGIS, Auth, Storage, Edge Functions)	**Free** through Sprint 3; **Pro** from Sprint 4 or whenever storage nears the limit	$0 → ~**$25**	Free plan: 2 projects, ~500 MB database, ~1 GB file storage, auto-pause after 1 week of inactivity (as of 2026 summaries; **VERIFY** on supabase.com/pricing). The deck's own maths (~3 GB/month at 500 reports/day) shows free storage is not enough for a busy pilot (**C-08**).
Firebase (Cloud Messaging, Crashlytics, App Distribution)	Free	$0	**VERIFY** current free-plan terms; keep Firebase to these three products so no billing plan is needed.
Google Maps Platform	Pay-as-you-go with **free monthly usage per product** (Essentials about 10,000 free calls per product per month; the old flat $200 credit ended in March 2025)	$0 expected; **budget $5–10** buffer	Set a quota cap per API and budget alerts at 50% and 80%. Billing must be enabled even when usage is free.
Dashboard hosting	Firebase Hosting free tier or Cloudflare Pages free tier	$0	Avoid hosting plans restricted to non-commercial use if the pilot is run with a municipality (**VERIFY** terms).
Domain name	One `.in` or `.app` domain	~$1–2	Optional until beta.
Model training	Google Colab free tier or university lab GPU	$0	Colab free GPU time is not guaranteed; keep checkpoints.
GitHub (repo, Actions, Pages, Dependabot)	Free	$0	Private-repo Actions minutes are limited (**VERIFY**); public repo is unlimited for standard runners.
Sentry or similar web error	Free developer tier	$0	Optional; Crashlytics covers the app.

Item	Plan	Expected monthly cost	Note
tracking			
Contingency	Buffer for overages	~$10	
Worst-case total		**≈ $45–48**	Under the $50 target. A single overage month on Maps would eat the buffer, hence the quotas.

Not in the budget (and why): SMS/OTP messages (rejected, **D-06**), Apple Developer Program for iOS (about $99 per year; iOS is a stretch goal, **A-09**), paid Figma Dev seat, paid MkDocs themes.

3. Layer: Flutter (citizen app and dashboard)

Type	Item	Purpose	Why it fits	Catch	Sprint
MCP	**Dart and Flutter MCP server** (ships with the Dart SDK, started with `dart mcp-server`; **VERIFY** minimum SDK version)	Lets an AI assistant analyse code, run tests, inspect widgets and look up package docs	Both apps are Flutter; official, no cost	Needs a recent Dart/Flutter SDK. Review any file changes before accepting.	1
MCP	**Figma MCP server** (Figma's Dev Mode server)	Brings frames and design tokens into code	Our UX Lead designs in Figma; helps keep `design.md` tokens and code in sync	Access to Dev Mode features depends on the Figma plan and seat (**VERIFY**); rate limits on free plans	1–2
MCP	**Context7** (Upstash)	Pulls current package documentation into the assistant	Flutter packages change quickly; avoids stale API guesses	Free tier with request limits (**VERIFY**)	1
IDE	**Flutter** and **Dart** extensions for VS Code, or Android Studio with the Flutter plugin	Run, hot reload, debug, widget inspector	Standard tools	None	1
IDE	**Claude Code** extension for VS Code (or another assistant)	AI pair-programming inside the editor	Team is small; speeds boilerplate	Requires a Claude plan; review all generated code	1
IDE	**Error Lens**, **Pubspec Assist**	Inline errors; add packages safely	Small quality-of-life gains		1

Type	Item	Purpose	Why it fits	Catch	Sprint
OSS	`camera` (Flutter team)	Camera preview and capture	Official plugin	Test on 3+ Android models; behaviour differs by phone	1
OSS	`geolocator` (Baseflow, MIT)	GPS reading, accuracy, mock-location flag on Android	Gives `accuracy` and `isMocked` needed by `rules.md` §2	Handle permission flows carefully	1
OSS	`google_maps_flutter` (Flutter team, BSD)	Maps in app and dashboard	Same SDK on mobile and web	Counts toward Maps free usage; web map loads count as Dynamic Maps	1, 4
OSS	`drift` (MIT) over SQLite	Offline queue and local cache	Typed queries, migrations, testable	Keep the queue schema versioned	2
OSS	`workmanager` (MIT)	Background sync on Android	Sends queued reports without opening the app	Background limits vary by phone maker; test on at least two brands	2
OSS	`connectivity_plus`, `flutter_image_compress`, `flutter_secure_storage`, `image`	Network state, WebP compression, token storage, image processing	Match TRD-P9 and `security.md` §4	Check each package's maintenance before adding	1–2

Type	Item	Purpose	Why it fits	Catch	Sprint
OSS	`firebase_messaging`, `firebase_crashlytics`	Push and crash reports	Free; needed for status notifications	Don't add other Firebase products by accident	4
OSS	`fl_chart` (MIT)	Dashboard analytics charts	Pure Flutter, light		4
OSS	`flutter_localizations` + `intl`	English and Hindi strings	Official	Hindi review needed (**Q-08**)	1

4. Layer: TensorFlow and TensorFlow Lite (AI)

Type	Item	Purpose	Why it fits	Catch	Sprint
MCP	**Hugging Face MCP server** (**VERIFY** availability)	Search models, datasets and papers from the assistant	Faster literature and dataset discovery	Dataset licences vary; record each one	1
MCP	**Context7** (see §3)	Current TensorFlow and Keras docs	TF/TFLite APIs and naming have changed (TensorFlow Lite is now also called LiteRT)		3
IDE	**Python**, **Jupyter** extensions (VS Code); **Colab** in the browser	Notebooks and scripts	Matches Colab workflow	Colab sessions can disconnect; save to Drive	1
IDE	**Netron** (VS Code extension or web)	Inspect `.tflite` graphs and quantization	Helps check INT8 conversion		3
IDE	**TensorBoard** (built into VS Code)	Training curves	Free		3
OSS	**TensorFlow** / **Keras** (Apache-2.0)	Training and conversion	Proposal's choice	Keep versions pinned between training and conversion	3
OSS	`tflite_flutter` (Apache-2.0)	Run `.tflite` in Flutter	Works offline on Android	Confirm it supports the Flutter version in use;	3

Type	Item	Purpose	Why it fits	Catch	Sprint
				check the LiteRT migration status (**VERIFY**)	
OSS	**RDD2022** dataset and `sekilab/RoadDamageDetector` (Arya et al.)	Public road-damage images, including an India subset	Cited in the Screen-1 deck	Labels are for *detection* (boxes), not whole-image classes; we need to crop or relabel. Check the dataset licence (**VERIFY**).	1
OSS	Other public pothole/road-damage datasets (for example on Kaggle or Roboflow Universe)	Extra pothole, drain and water images	Fills class gaps	Licences differ widely; some are non-commercial or unclear. Record each.	1
OSS	**Label Studio** (Apache-2.0) or **CVAT** (MIT)	Label field images	Free, self-hosted or local	Needs a laptop to host; export in a standard format	1
OSS	**Albumentations** (MIT)	Image augmentation	Helps with small data		3
OSS	**DVC** (Apache-2.0)	Version datasets and models	Reproducible training without large files in Git	Needs a storage remote (a shared Drive folder works)	3

Type	Item	Purpose	Why it fits	Catch	Sprint
OSS	**TensorFlow Lite Model Maker**	(Avoid)	Appears in old tutorials	No longer maintained; use plain Keras transfer learning instead (**VERIFY**)	n/a
OSS	**Ultralytics YOLO**	(Possible v1.1 detector)	Strong detection	**AGPL-3.0**: copyleft obligations apply to distributed apps; consider TensorFlow-based detectors instead	post-pilot

5. Layer: Supabase and Firebase (cloud)

Type	Item	Purpose	Why it fits	Catch	Sprint
MCP	**Supabase MCP server** (official, open-source)	Let an assistant inspect schema, draft migrations, run read queries	Postgres and RLS are central to the project	**Dev project only, read-only scope, fake data.** A model that can run SQL on real data is a risk (T16 in `security.md`). Don't connect it to production.	2
MCP	**Firebase MCP server** (included in the Firebase CLI; **VERIFY**)	Manage Firebase project settings and look up docs	Used only for push, crash and hosting	Same caution: least privilege	4
MCP	**GitHub MCP server** (official)	Issues, pull requests, Actions status from the assistant	Weekly sprint reviews and tracking	Use a fine-grained token limited to our repo	1
IDE	**PostgreSQL** extension for VS Code; **Supabase CLI**	Browse the database; run local Supabase with migrations	Local-first development avoids burning free-tier quota	Docker is needed for the local stack	2
IDE	**SQL formatter / linter** (e.g. `sqlfluff`, MIT)	Keep migrations tidy			2
OSS	**Supabase** (Apache-2.0) and `supabase_flutter` (MIT)	Auth, DB, storage, realtime	Postgres + PostGIS + RLS fit our needs (**D-02**)	Free plan pauses after a week idle; Pro plan from Sprint 4	2
OSS	**PostGIS** (GPL-2.0-or-later, used as a database extension)	`ST_DWithin`, point-in-polygon, `ST_SnapToGrid`	Needed for ward lookup, clustering and density cells	Used as a hosted extension, not redistributed by us	2

Type	Item	Purpose	Why it fits	Catch	Sprint
OSS	`pg_cron`	Scheduled SLA checks and retention jobs	Avoids a separate server	Check it is enabled on the chosen plan (**VERIFY**)	4
OSS	**pgTAP** (via `supabase test db`)	Database tests, including RLS	Keeps `security.md` §10 honest		2
OSS	**Firebase Cloud Messaging**, **Crashlytics**, **App Distribution**	Push, crash logs, beta APKs	Free; simple	Adding Cloud Functions or Cloud Storage may require a billing plan; we avoid them (**VERIFY**)	4–5

6. Layer: Google Maps Platform and map data

Type	Item	Purpose	Why it fits	Catch	Sprint
MCP	**Google Maps Platform Code Assist / developer-docs MCP** (**VERIFY** availability and name)	Up-to-date Maps API documentation and samples for assistants	Maps APIs change often	Treat generated code as a draft	1
IDE	**Google Cloud Code** or the Cloud Console (browser)	Manage keys, quotas, budgets	Needed to set quotas and alerts	Don't leave keys unrestricted	2
OSS/API	**Maps SDK for Android and Maps JavaScript API** (through `google_maps_flutter`)	Display maps	Proposal's choice	Free usage is per product and doesn't pool; set quotas. **HeatmapLayer is no longer available (May 2026)**, so we draw our own density cells (`architecture.md` §7).	1, 4
API	**Geocoding API**	Optional human-readable address label	Helpful on the ticket card	Don't store results long-term; we use our own PostGIS wards for logic (**D-08**)	4
OSS	**OpenStreetMap data** (via Overpass queries or a regional extract; ODbL)	Roads, schools, hospitals, markets for `context_features`	Free and good coverage in many Indian cities (**VERIFY** for Agra and Jodhpur)	Credit OSM; follow share-alike for derived databases; don't hammer public servers	2–4
OSS	`flutter_map` (BSD-3)	Fallback map if Maps quota is exhausted	Free	OSM's public tile servers are not for	4

Type	Item	Purpose	Why it fits	Catch	Sprint
				heavy use; use a tile provider with a free tier or self-host (**VERIFY** terms)	
Data	**Ward boundaries:** city corporation open data, OSM admin boundaries, or community datasets (for example from the DataMeet community; **VERIFY** coverage and licence)	`wards.geom`	Needed for routing and permissions	Quality and recency vary; may need manual digitising (**Q-04**)	2

7. Layer: CI/CD

Type	Item	Purpose	Why it fits	Catch	Sprint
MCP	**GitHub MCP server** (see §5)	Pull request and workflow status			1
IDE	**GitHub Actions** and **GitHub Pull Requests** extensions	See CI results in the editor			1
OSS/Service	**GitHub Actions**	Run tests, build APKs, apply Supabase migrations	Free for public repos; limited minutes for private (**VERIFY**)	Keep builds short; cache dependencies	1
OSS	`subosito/flutter-action`	Install Flutter in CI	Widely used	Pin the Flutter version	1
OSS	`supabase/setup-cli` action	Run migrations and database tests in CI		Needs project access tokens in CI secrets	2
OSS	**Fastlane** (MIT)	Build and sign Android releases	Repeatable releases	iOS signing needs an Apple account	5
Service	**Firebase App Distribution**	Send beta APKs to testers	Free	Invite testers by email	5
OSS	**Dependabot**, **Gitleaks** (MIT)	Dependency updates, secret scanning	Supports `security.md` §8		1

Type	Item	Purpose	Why it fits	Catch	Sprint
Service	Codemagic or similar for iOS builds	Build iOS without a Mac	Free allowance exists (**VERIFY**)	Still needs an Apple Developer account	stretch

8. Layer: Testing

Type	Item	Purpose	Why it fits	Catch	Sprint
MCP	**Playwright MCP server** (Microsoft)	Drive a browser from an assistant to test the dashboard	Dashboard flows (verify, assign, resolve)	Runs against a dev dashboard only	4
MCP	**Sentry MCP** (**VERIFY**)	Pull error details into the assistant	Only if Sentry is used		5
IDE	**Flutter test** runner, **Test Explorer**	Run unit and widget tests			1
OSS	`flutter_test`, `integration_test`, `mocktail` (MIT)	Unit, widget and device tests	Official or standard		1
OSS	**Patrol** (Apache-2.0; **VERIFY**)	Test native permission dialogs (camera, location)	Our core flow depends on them	Slightly heavier setup	2
OSS	**pgTAP** (see §5)	Row-level-security and rule tests			2
OSS	**pytest** (MIT)	Model code and conversion tests			3
OSS	**Playwright** (Apache-2.0)	Dashboard end-to-end tests			4
OSS	**Locust** (MIT)	Load script for the 100× burst test (TRD-P10)	Simple Python	**k6** is a good tool but AGPL-3.0; we prefer Locust	4
OSS	**OWASP ZAP** (Apache-2.0)	Baseline web security scan	Supports `security.md` §8		5

Type	Item	Purpose	Why it fits	Catch	Sprint
Process	Real-device matrix: 3 Android phones (low, mid, higher)	Camera, GPS and inference timing	Emulators lie about these	Collect devices early	1

9. Layer: Documentation

Type	Item	Purpose	Why it fits	Catch	Sprint
MCP	**GitHub MCP** (docs PRs) and **Context7**	Review and look up			1
IDE	**Markdown Preview Mermaid Support**, **markdownlint**, **Mermaid Chart** (VS Code)	Preview and lint these files and their diagrams	The set uses Mermaid		1
IDE	**Vale** (MIT) with a plain-language style	Flag jargon and long sentences in user-facing text and docs	Backs the "warm, plain" voice requirement	Needs a style config; start with a short rule set	1
OSS	`@mermaid-js/mermaid-cli` (MIT)	Check that diagrams render, in CI	Stops broken diagrams reaching judges	Needs a headless browser in CI	1
OSS	`lychee` link checker (MIT/Apache)	Catch dead links			1
OSS	**MADR** decision-record template (MIT)	Format for entries in `memory.md`	Light and familiar		1
OSS	**MkDocs** (BSD) with the Material theme, or **Docusaurus** (MIT)	Publish the docs set as a site	Free hosting on GitHub Pages	Check the theme's maintenance status before committing (**VERIFY**)	5

Type	Item	Purpose	Why it fits	Catch	Sprint
Reference	**mySociety FixMyStreet** (AGPL-3.0)	Study how a mature reporting platform handles nearby-report prompts and workflow	Real-world civic reporting UX	**AGPL**: study ideas; don't copy code into CivicSync	1

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10. Using MCP servers and AI assistants safely

MCP servers give an AI assistant hands. That is useful and risky. For this project:

1. **Dev only.** Connect assistants to a dev Supabase project filled with fake data. Never connect production.
2. **Smallest scope.** Use read-only mode where the server offers it; use fine-grained tokens (for example GitHub tokens limited to this repository).
3. **No secrets in prompts or files** that an assistant can read. The service-role key is never on a developer laptop.
4. **Database text is data.** Citizen notes could contain instructions. Nothing the assistant reads from a table is a command (`security.md`, T16).
5. **Review everything.** Every AI-written change goes through the same pull-request review as human code.
6. **Log it.** Note which assistant and server were used for a given change in the pull request description.
7. **New servers need a second pair of eyes.** Check who publishes it, what permissions it asks for, and when it was last updated.

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11. Licence flags in one place

Item	Licence	What to do
Ultralytics YOLO	AGPL-3.0	Avoid in the shipped app unless we accept AGPL terms
k6	AGPL-3.0	Use only as an internal test tool, or use Locust
mySociety FixMyStreet	AGPL-3.0	Reference only; no code copying
PostGIS	GPL-2.0-or-later	Used as a service-side extension; not redistributed
OpenStreetMap-derived data	ODbL	Credit; share-alike for derived databases
Public datasets (RDD2022, Kaggle, Roboflow)	Varies	Record each licence in model release notes; avoid non-commercial-only sets if the product may be commercial
Noto Sans / Noto Sans Devanagari	OFL	Free to bundle
Google Maps	Terms of service	Don't cache or store results outside the terms; attribution required

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12. Document acceptance criteria

This document passes review when it meets the checklist for `tooling.md` in `loop.md` §3.